

NAMD & VMD

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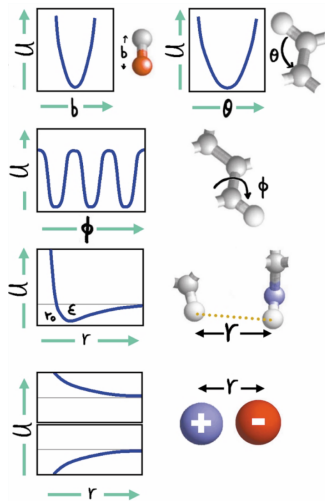
NAMD is a molecular dynamics (MD) simulation engine

- Generic research question: *How does this molecular system behave as a function of time?*
- Requirements:
 - Empirical model of the interactions between atoms == **force field**
⇒ We get the energies and can calculate the forces acting on each atom and integrate Newton's equations of motion for the time evolution
- Equilibrium and non-equilibrium processes
- NAMD is a one implementation developed by researchers at the University of Illinois Urbana-Champaign

Force field

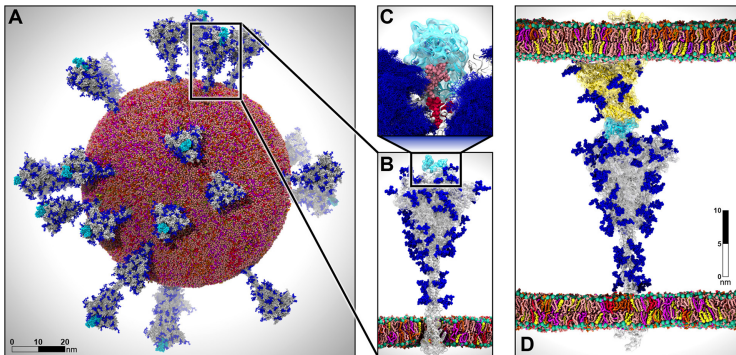
$$\begin{aligned}
 U = & \frac{1}{2} \sum_{\text{bonds}} k_r (r - r_0)^2 + \frac{1}{2} \sum_{\text{angles}} k_\theta (\theta - \theta_0)^2 \\
 & + \sum_{\text{dihedrals}} k_\phi [1 + \cos(n\phi - \gamma)] \\
 & + \sum_{\text{vdW}} 4\varepsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] \\
 & + \sum_{\text{electrostatics}} \frac{q_i q_j}{r_{ij}}
 \end{aligned}$$

- Bonded and non-bonded terms
- System-specific (proteins, lipids, DNA, ...)



NAMD used in two Gordon Bell award-winning projects

- 305 million atoms, 68 ns/day on >24 000 Nvidia V100 GPUs (Summit supercomputer, ORNL)



Implementation details

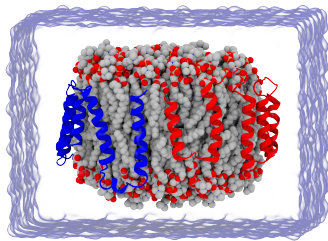
- Written in C++ leveraging an in-house parallel programming framework **Charm++** (not to be confused with CHARMM)
 - `charmplusplus.org`
- Variety of build options
 - Single-node: Serial and multicore
 - Multi-node: MPI and SMP (analogous to hybrid MPI/OpenMP parallelism)
 - GNU compiler stack and OpenMPI recommended
 - GPU: CUDA and HIP/ROCm support
- Uniform domain decomposition into “patches”, dynamic load balancer assigns patches to processors, monitors the load and adjusts as necessary

NAMD input files and parameters

- 4 files are required:
 1. Force field file (CHARMM or X-PLOR format)
 2. NAMD configuration file (simulation parameters)
 3. Molecular structure / topology (PSF file)
 4. Starting coordinates (PDB file)
- NAMD configuration file requires *at least*:
 1. numsteps
 2. coordinates
 3. structure
 4. parameters
 5. exclude
 6. outputname
 7. One of either: temperature, velocities or binvelocities

ApoA1

- This exercise uses *Apolipoprotein A1* as an example system
 - Found in blood plasma, participates in lipid metabolism by acting as an enzyme co-factor and cell membrane receptor ligand
 - Used as a biomarker – a lowered ApoA1 concentration correlates with an increased risk of cardiovascular diseases
- A representative example of a biomolecular system, often used as a (small) benchmark case for MD codes (92k atoms)



Let's look at the ApoA1 configuration file

- Notes and illustrations on computing non-bonded interactions:
 - <https://www.ks.uiuc.edu/Research/namd/3.0/ug/node25.html>
 - <https://www.ks.uiuc.edu/Research/namd/3.0/ug/node89.html>

Performance considerations

- NAMD performance typically reported as ns/day (or days/ns)
 - `grep "Benchmark" <output filename>`
- Don't run too short benchmarks (>1000 steps ideally)
 - You may also use `benchmarkTime <value>` to set wall-time instead
- Mind your I/O, do not output data every timestep!
 - For example, `outputEnergies` is 1 by default(!)
- Computing non-bonded interactions is heavy, decreasing cutoff may improve performance, but decreases accuracy

Performance considerations

- SMP builds have a smaller per-core memory footprint, but requires you to leave one thread per process for communication
 - `srun namd3 +ppn $((SLURM_CPUS_PER_TASK - 1)) <config file>`
 - Good to compare performance with non-SMP builds
- Large systems (>100k particles) run efficiently on high-end GPUs
 - GPU-resident mode is beneficial (everything calculated on the GPU, no host-device data transfer)
- See the user guide for more details:
 - <https://www.ks.uiuc.edu/Research/namd/3.0/ug/node87.html>
 - <https://www.ks.uiuc.edu/Research/namd/3.0/ug/node101.html>
 - <https://www.ks.uiuc.edu/Research/namd/3.0/ug/node106.html>

VMD: a visualization program developed by NAMD team

- High-quality rendering and also tools for analysis
- Typical use case:
 1. We want to view a static structure or MD trajectory
 2. Then render a nice picture (or movie) for our publication
- Output trajectory with for example:
 - `DCDfile apoa1.traj`
 - `DCDfreq 1000`
- "Graphics" > "Representations" and "Display" are the main views for tuning the visualization
- A brief tutorial: https://a3s.fi/sscc/VMD_rendering.tgz

More information

- NAMD user guide
 - <https://www.ks.uiuc.edu/Research/namd/current/ug/>
- VMD user guide
 - <https://www.ks.uiuc.edu/Research/vmd/current/ug/>
- NAMD documentation in Docs CSC
 - <https://docs.csc.fi/apps/namd/>